

LT spice simulation of the inductor coil-

<https://wiki.analog.com/university/courses/electronics/electronics-lab-speaker>
[https://eng.libretexts.org/Bookshelves/Electrical_Engineering/Electronics/Laboratory Manual - AC Electrical Circuit Analysis \(Fiore\)/17%3A Loudspeaker Impedance Model/17.5%3A Procedure](https://eng.libretexts.org/Bookshelves/Electrical_Engineering/Electronics/Laboratory_Manual_-_AC_Electrical_Circuit_Analysis_(Fiore)/17%3A_Loudspeaker_Impedance_Model/17.5%3A_Procedure)

We will use the simulation of a loudspeaker as the base of our comparative analysis.

The moving system of the loudspeaker, including the cone, cone suspension, spider and the voice coil, has a certain mass and compliance. This is most commonly modeled as a simple mass suspended by a spring that has a certain resonant frequency at which the system will vibrate most freely.

This frequency is known as the “free-space resonance” of the speaker and is designated by F_S . At this frequency, since the voice coil is vibrating with the maximum peak-to-peak amplitude and velocity, the back-emf generated by coil motion in a magnetic field is also at its maximum. This causes the effective electrical impedance of the speaker to be at its maximum at F_S , known as Z_{MAX} . For frequencies just below resonance, the impedance rises rapidly as the frequency approaches F_S and is inductive in nature. At resonance, the impedance is purely resistive and beyond it, as the impedance drops, it looks capacitive. The impedance reaches a minimum value, Z_{MIN} , at some frequency where the behavior is mostly (but not perfectly) resistive over some range of frequencies. A speaker's rated or nominal impedance, Z_{NOM} , is derived from this Z_{MIN} value.

Dynamic loudspeakers are abysmally inefficient electro-mechanical conversion devices

We will model our voice coil according to the following-

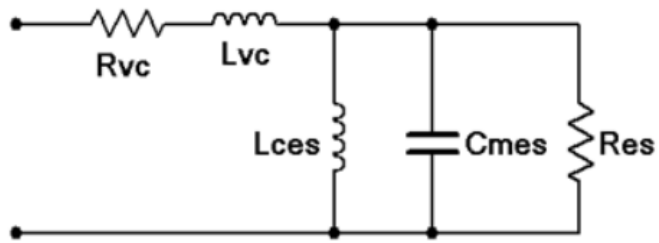


Figure 17.4.1

Here R_{vc} and L_{vc} refer to the resistances and inductances that are native to the voice coil.

L_{ces} refers to the “springiness” of the medium (driver compliance), C_{mes} refers to the driver's mass while R_{es} refers to the

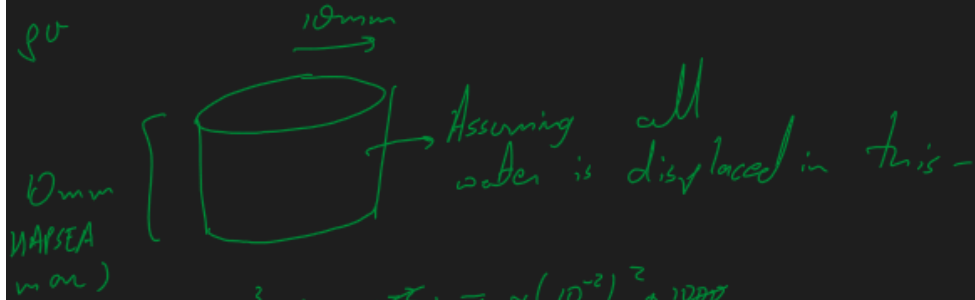
Calculations-

The 20mm coil and magnet were used for the calculations

#1 → Calculation Moving mass of the actuator -
 = mass of magnet + mass of displaced water

$$m = 7.1 + m \dots \text{water}$$

PO



MAPSEA
 m m)

$$= 7.1 \times 10^{-3} + 1.2 \times 10^{-2} \times \pi \times (10^{-2})^2 \times 1000$$

$$= 7.1 \times 10^{-3} + 3.74 \times 10^{-3} = 10.2 \times 10^{-3}$$

$$= 10.2 \text{ mN}$$

#2 Calculating the force constant of the device -

$$\text{using } 0.1 \text{ N} \Rightarrow 0.1 \text{ mm}$$

$$k = C = x/f \rightarrow \frac{10^{-4}}{10^{-4}} = \frac{10^{-3}}{1} \text{ F} = 1 \text{ mF}$$

#3 calculating the resistance to the magnet -

$$R_{\text{rod}} \approx f \dots$$

$$\approx 0.64$$

Analysis-

Using the voltage between the inductor and the capacitor, we can consider the voltage here to be directly proportional to the displacement of our coil-

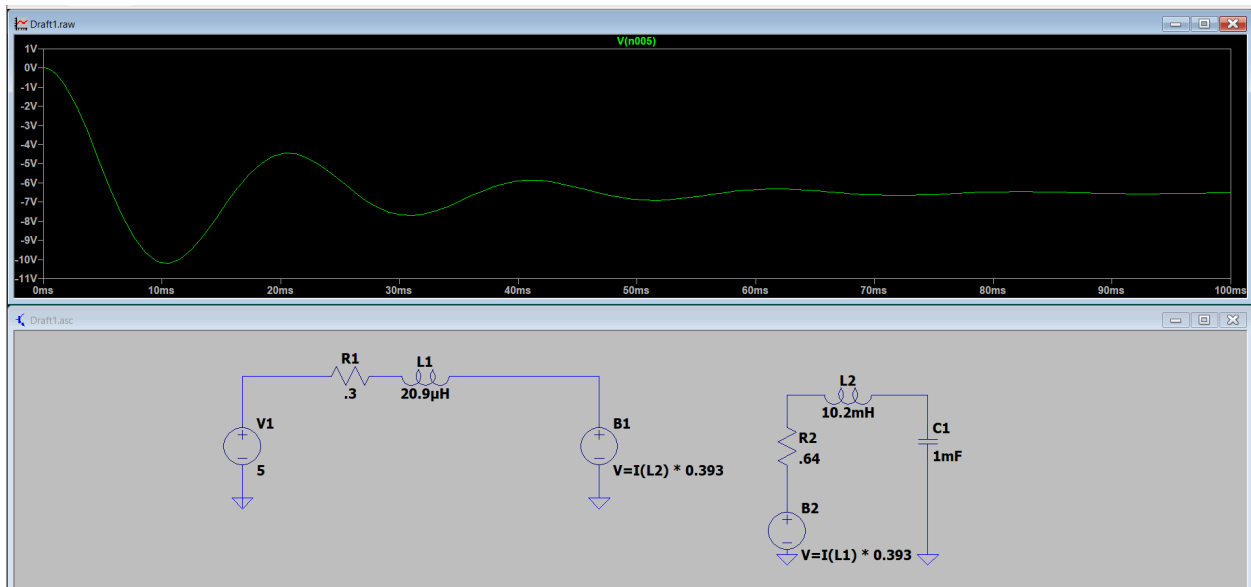
Case 1 (5V amplitude, step voltage)

Max displacement = -10

Time for max displacement = 10ms

Steady state = -6.52

2% settling time = 45ms



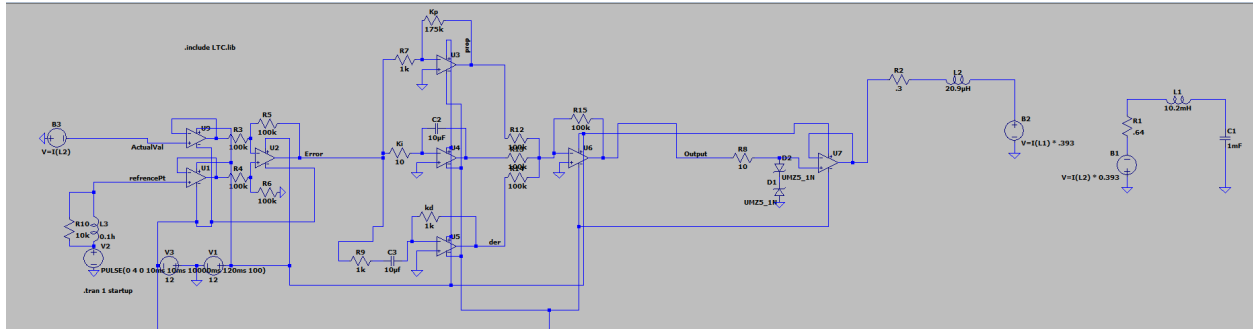
Case 2 (12V amplitude, step voltage)

It achieves 6.2 within 3.23ms (without control logic)

To get a better idea on how the voltage affects the rate of change, we will use a simple analog PID loop to control the values and set it to be maximum at 5V

Using a PID controller-

An analog PID controller was added to get a better understand of how different voltages will affect the movement of the coil. In these diagrams V(C1) is analogous to the displacement of the magnet (considering that the voice coil is fixed).



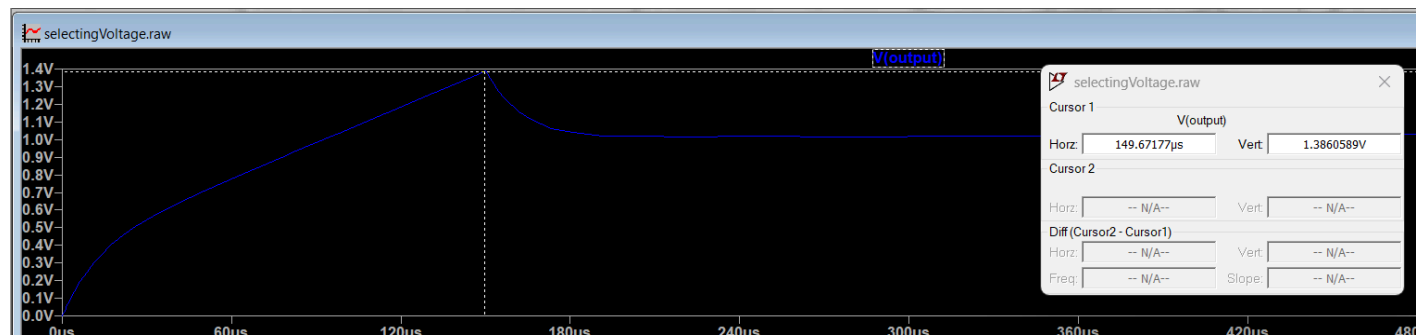
The controller aims to control the current to ensure that there is a constant current 4A through the coil. This directly translates to how the magnet will be deflected. Zeiner diodes have been placed to study different maximum voltages (5V and 12V)

The PID controller has been tuned to the best of my ability. It is possible to achieve a better response.

Results and analysis-

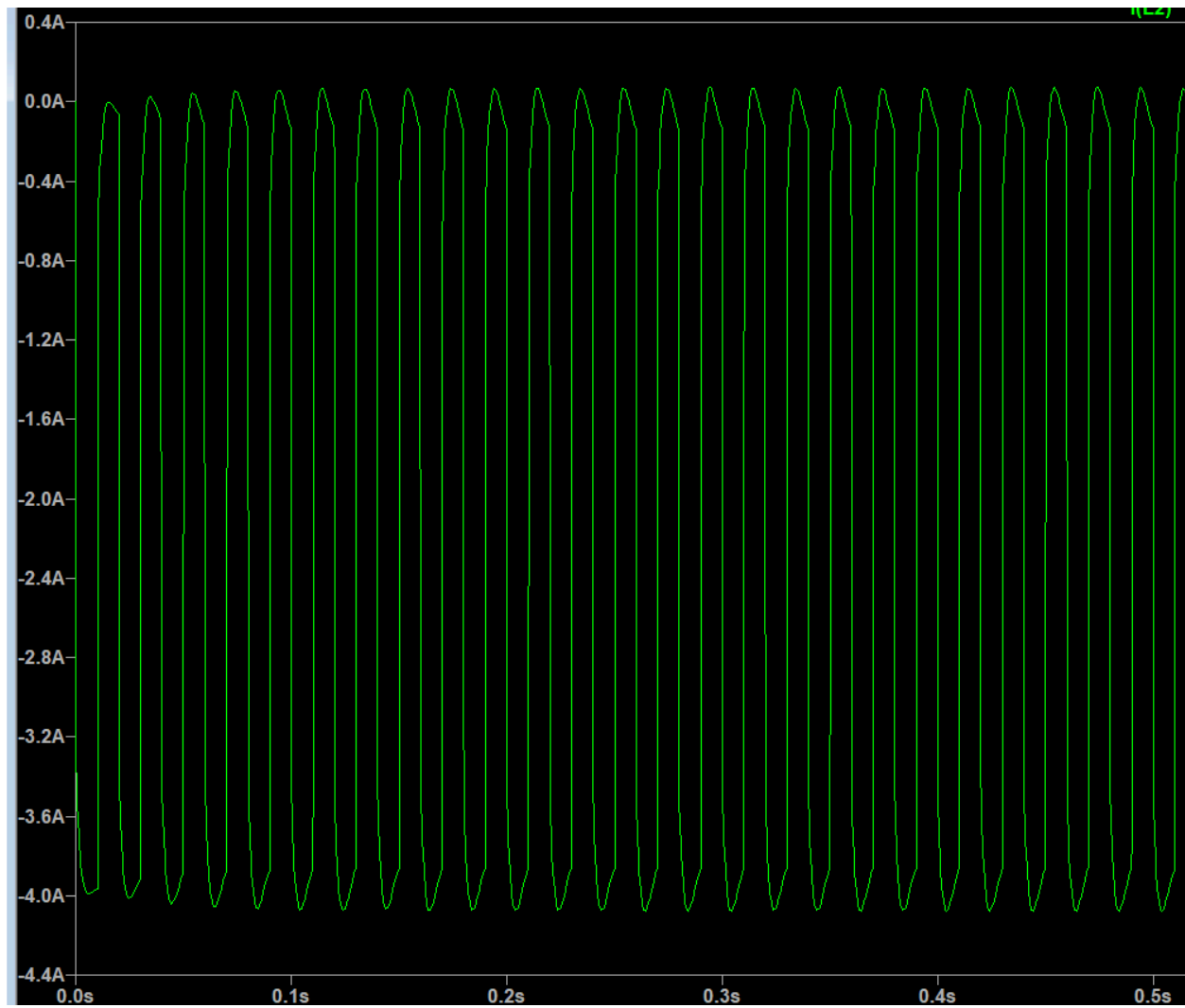


The current reaches a steady state value of 4A within around 3ms (2% settling time)

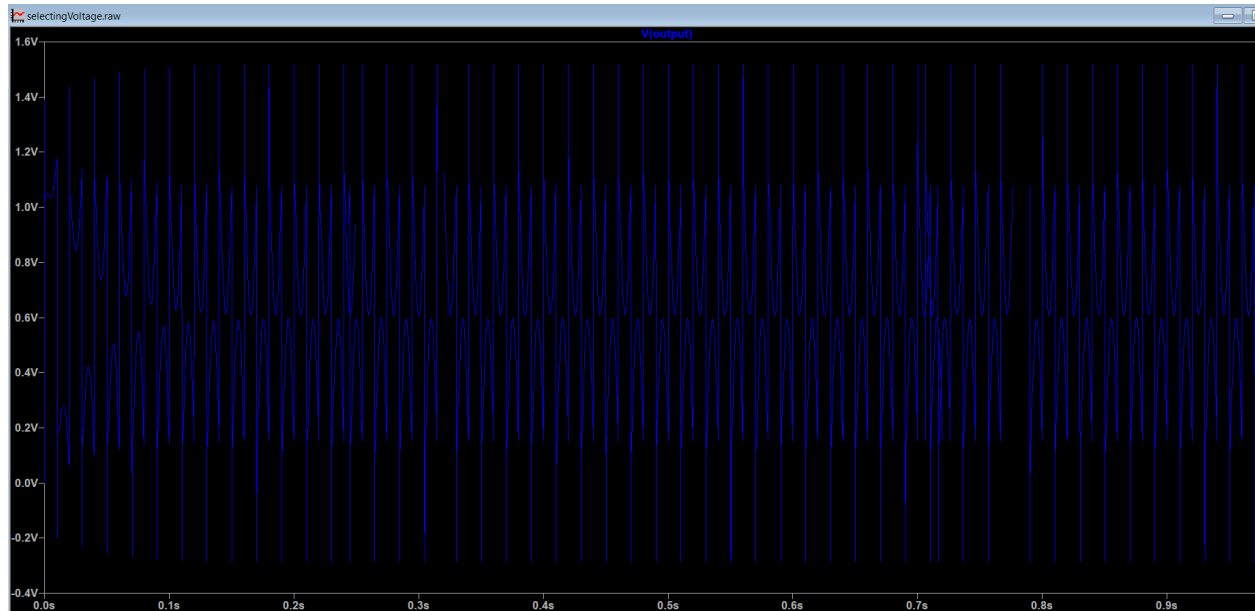


The maximum voltage that is observed is around 1.4V

The maximum theoretical frequency is around 150Hz

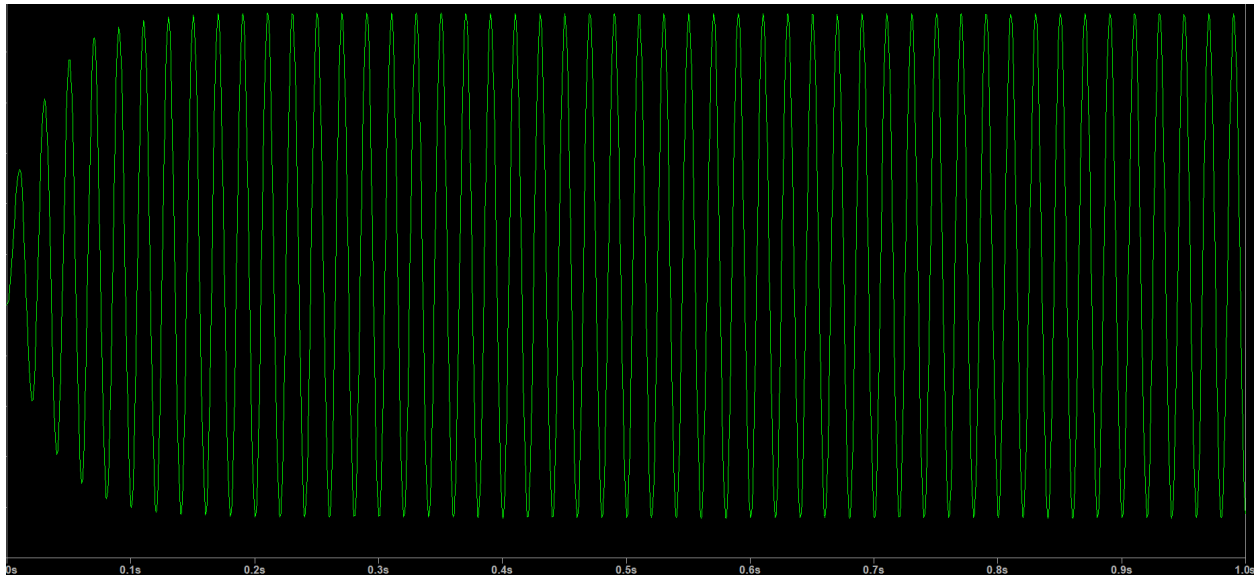


Output at 50 Hz



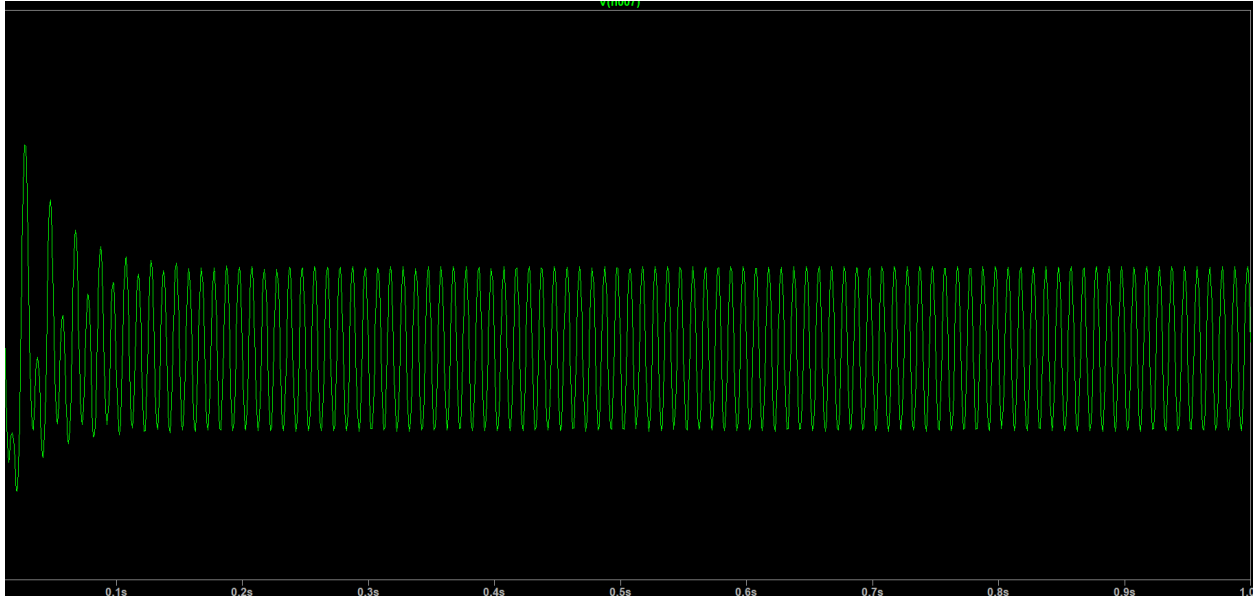
Voltage spikes at 50Hz

Max spike = 1.5V, -310mv

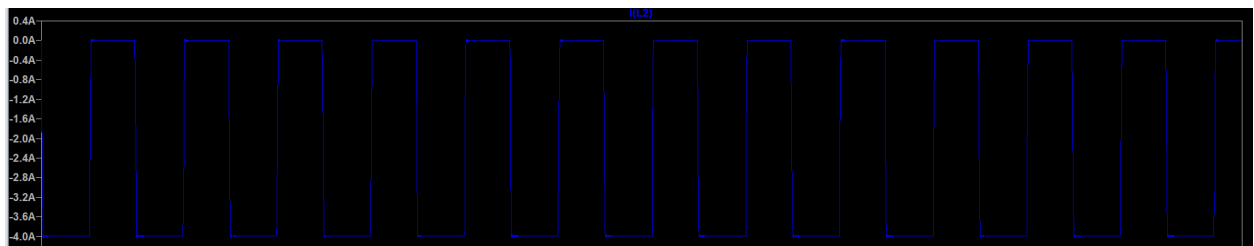


Voice coil response

**Trying a higher frequency-
(100Hz)**



Voice coil deflection - we notice a decrease in deflection of the voice coil (4x decrease)



Since maximum current has been achieved, there is not much scope for improving the deflection of the voice coil. It has been limited by the nature of the voice coil as well as the actuator. It is possible to get more deflection by passing more current, but this may lead to the overheating of the coil and burning out of the insulation leading to a short circuit.

References-

PID -

[bing.com/ck/a?!&&p=1301bd96e7b9ff1549c602d2168864cb39d600987e7c0a48434b91b3133031c7JmltdHM9MTc4MjE3MjgwMA&ptn=3&ver=2&hsh=4&fclid=1b781d3a-68c6-6eab-2eff-080369346f84&psq=Design+of+Analog+PID+Controller+journal&u=a1aHR0cHM6Ly96ZW5vZG8ub3JnL3JlY29yZHMvMTQ2NzQ2MC9maWxlcy9KQUVUVjdJMTAyLnBkZg&ntb=1](https://www.bing.com/ck/a?!&&p=1301bd96e7b9ff1549c602d2168864cb39d600987e7c0a48434b91b3133031c7JmltdHM9MTc4MjE3MjgwMA&ptn=3&ver=2&hsh=4&fclid=1b781d3a-68c6-6eab-2eff-080369346f84&psq=Design+of+Analog+PID+Controller+journal&u=a1aHR0cHM6Ly96ZW5vZG8ub3JnL3JlY29yZHMvMTQ2NzQ2MC9maWxlcy9KQUVUVjdJMTAyLnBkZg&ntb=1)

Converting mechanical to electrical-

Using a better magnet model-

Using the same formulas to model the magnetic force as in the HAPSEA paper, we get the following-

Magnetic forces -

Hapsea paper -

The coil is separated into infinitesimally thin loops

Biot savart + Lorentz force $\rightarrow$

$$m = \frac{4q_1 q_2}{(q_1 + q_2)^2 + x^2}$$

$q_1, q_2 \rightarrow$ radii of two specific thin loops

x = distance b/w them

Since the coil & magnet has a discrete thickness $\rightarrow$

$$r(nr) = r_c + \frac{nr-1}{Nr-1}(R_c - r_c)$$

r_c = inner rad

R_c = outer rad

nr = specific layer index

Now, through the principle of super position -

$$\sum_{n_m=1}^{N_m} \sum_{n_r=1}^{N_r} \sum_{n_\theta=1}^{N_\theta} F_j(r/n_r, R_m, r+2(n_r, n_\theta))$$

N_m = no of layers in the magnet.

N_r = no of radial layers in the coil

N_θ = no. of axial layers in the coil

Outer dia of the magnet.

magnetic field of a current carrying coil.

$$B = \frac{\mu_0 N I R^2}{2 (R^2 + x^2)^{3/2}}$$

$$= \frac{(4\pi \times 10^{-7}) 50 (10^{-2})^2 I}{2 (10^{-2} + (1.5 \times 10^{-2})^2)^{3/2}}$$

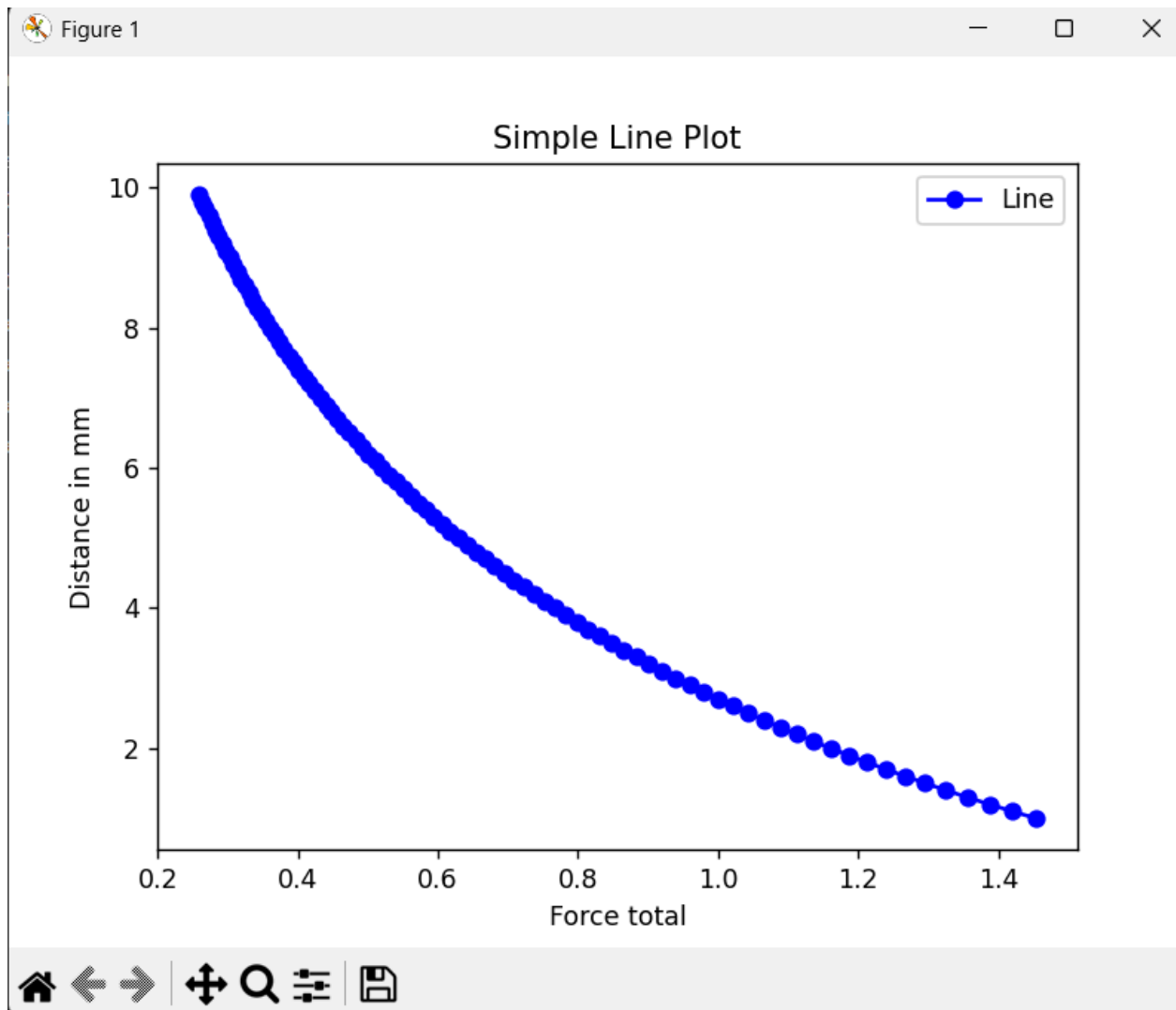
$$0.32 = \frac{6.28 \times 10^{-7} \times 50 \times 10^{-4} I}{2.06 \times 10^{-6}}$$

$$0.32 = \frac{628 \times 10^{-4} I}{2.06}$$

$$\frac{0.6592 \times 10^4}{62.8} = I$$

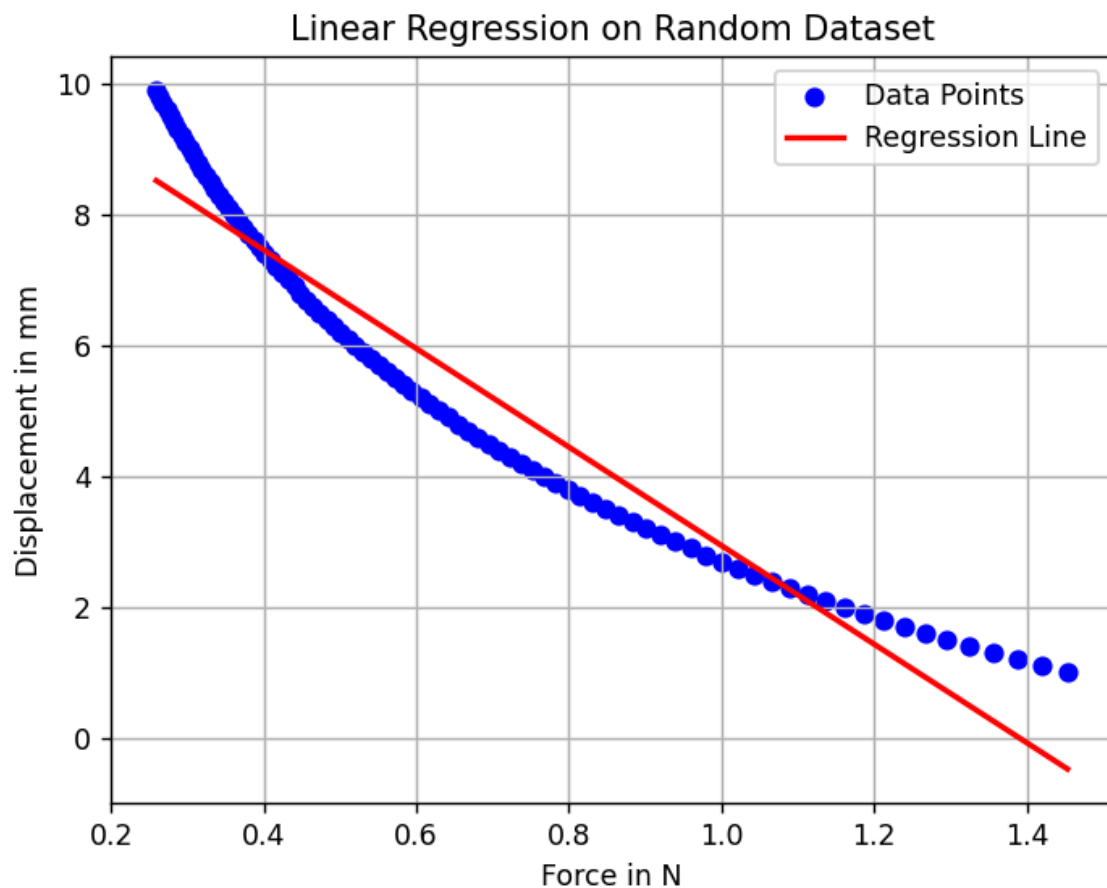
$$106.472$$

Using these equations, we get the following graph



Since the equations are nonlinear and since it would be difficult to model in LT Spice, for the time being, I have used linear regression to make an approximate model of this-

Figure 1



Using this data, we get the following formula for the line-

$$\frac{1}{5} \left(\frac{18.63 - 1000y}{7.47} \right) = x$$

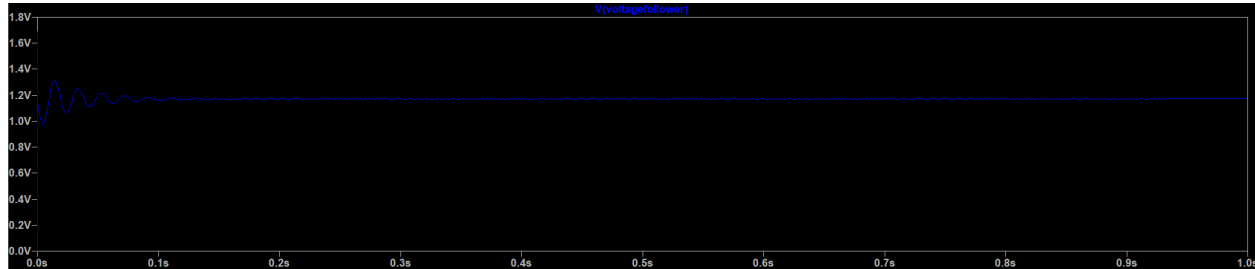
Here y is in mm, and x is in n. The $\frac{1}{5}$ factor was added as the original equation to correct for current. (we passed 5A in the current coil in this example)

Bringing this back into LT Spice, we get the following observations-

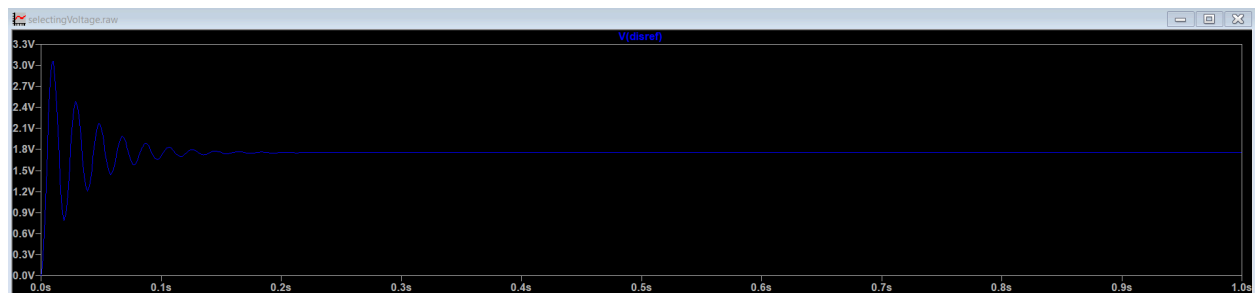
Note- in all cases, we are trying to achieve the maximum current in our voice coil as soon as possible as this is the only measurable parameter we have at the moment (current

through voice coil) and the only limiting parameter as well (too much current might cause the voice coil to overheat and short circuit)

Using a step impulse, we have-

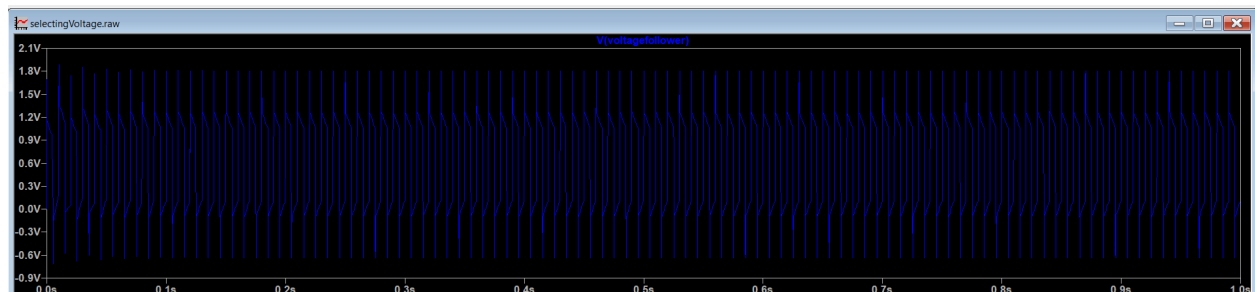


The maximum voltage that was observed across the voice coil is 1.7V.

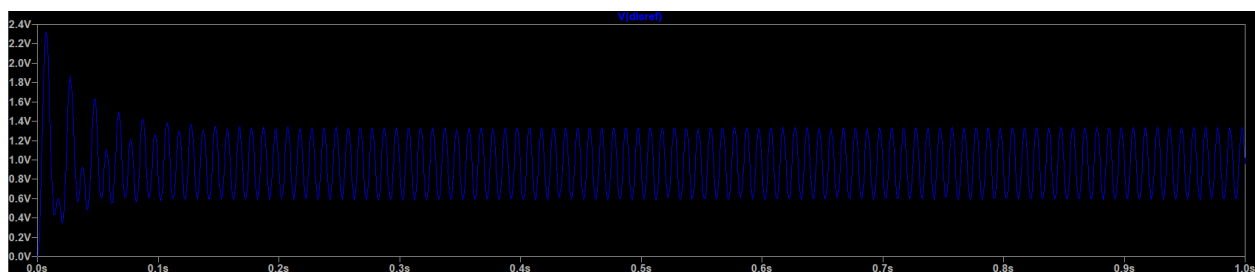


We have a maximum of 3.12mm displacement of the voice coil, while it settles down to 1.755mm after around 1ms

Using the 100Hz input frequency, we have-

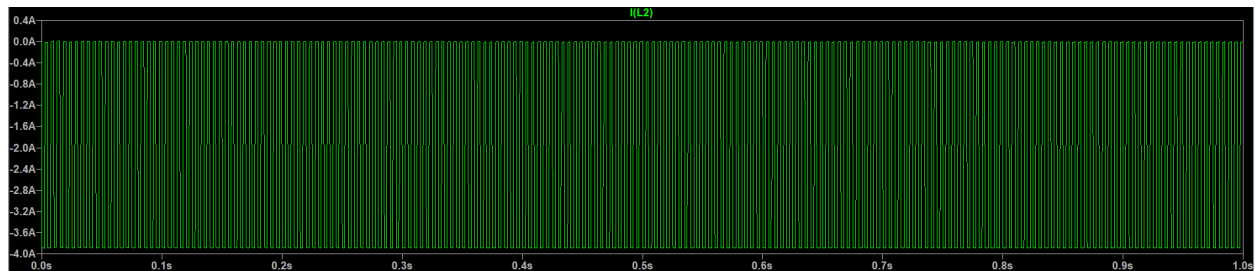


Max voltage across the coil is around 1.85V and -0.73V

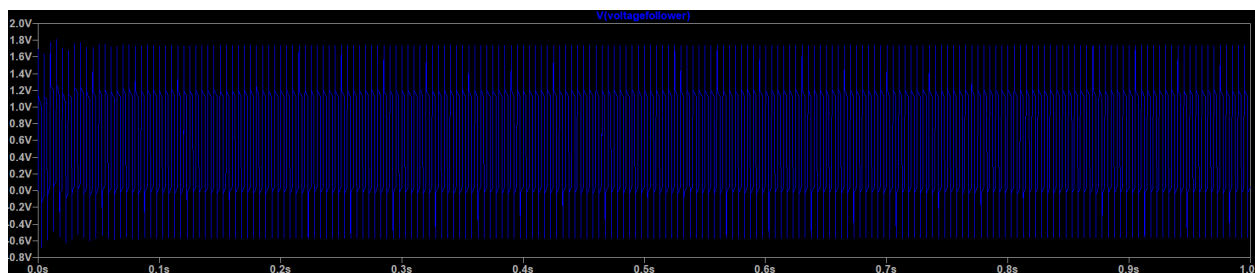


The maximum displacement of the magnet is 2.32mm, while it becomes steady at an amplitude of around $\pm 0.36\text{mm}$

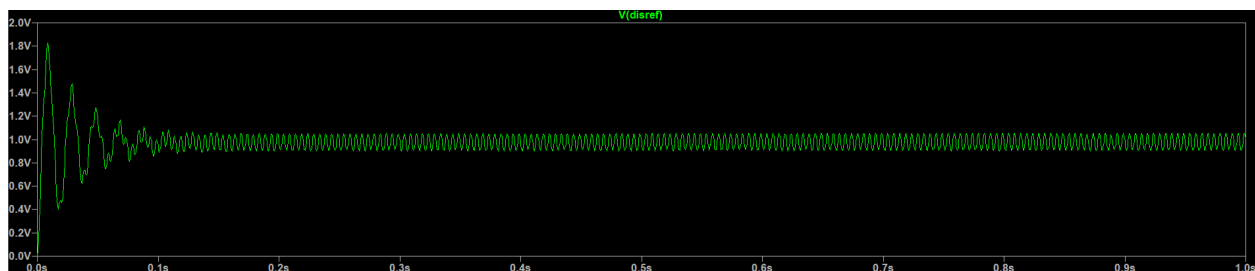
Trying with 200Hz-



The current across the coil seems well controlled at 200Hz, achieving its reference of 4A.



The maximum voltage seems to hover around 1.85V



The maximum displacement that we observe is 1.82mm, eventually settling down to an amplitude of 0.0733mm (peak to peak of 0.14mm)